

Harshit Kumar Sandhu

CONTACT INFORMATION	112, Department of Aerospace Engineering Indian Institute of Science, Bangalore Karnataka, India - 560012	<i>Email:</i> harshithk@iisc.ac.in <i>Mobile:</i> +91-9039000929
RESEARCH INTERESTS	• Solid Mechanics • Vibrations and Wave Propagation • Lattice Dynamics and Mechanical Metamaterials • Non-Hermitian Systems and Phononic Time Crystals	
EDUCATION	Indian Institute of Science, Bengaluru, Karnataka, India Doctor of Philosophy in Aerospace Engineering Jan 2020 – Present Grade Point Average: 8.6/10 • Tentative thesis topic: “Wave Propagation in Non-Hermitian Phononic Time Crystals” • Advisor: Prof. Rajesh Chaunsali Government Engineering College, Raipur, Chhattisgarh, India Bachelor of Engineering in Mechanical Engineering June 2016 Grade Point Average: 7.99/10 • Thesis topic: “Design and Simulation of a Hood Assembly for an Electric Mini-Crane” • Advisor: Lecturer Shashi Chichkhede	
WORK EXPERIENCE	THE GATE ACADEMY Pvt. Ltd., Bangalore Subject Matter Expert Aug 2019 – Dec 2019 THE GATE ACADEMY Pvt. Ltd., Bangalore Faculty Sep 2018 – July 2019 • Trained over 2000 GATE aspirants. • Developed course material for the training program. • Conducted training for students from state engineering colleges (under NPIU) for GATE preparation. Steel Authority of India Ltd. (SAIL), Bhilai, Chhattisgarh Vocational Trainee Aug 2014 • Gained exposure to various workshops at BSP-SAIL, including Blast Furnace, Coke Oven Battery, Steel Melting Shop, Machining Shop, and Rolling Mill. • Learned about the operations, functionality, and fundamental processes at each workshop.	
HONORS AND AWARDS	• AIR 53 in GATE Engineering Sciences (XE), 2019 – out of 3,559 candidates (98.5 percentile) • State Rank 625 in the state-level Chhattisgarh Pre-engineering Test (2012)	
PAPERS IN PREPARATION	[3] A. Umashankar, H. K. Sandhu & R. Chaunsali, <i>Characterizing a Parametric Oscillator using SINDy</i>. [2] H. K. Sandhu, A. Umashankar & R. Chaunsali, <i>Emulating Topological Edge Modes of a Phononic Time Crystal with a Single Parametric Oscillator</i>.	

	[1] H. K. Sandhu , S. Dutta & R. Chaunsali, <i>Wave Propagation in an Elastic Lattice with Nonreciprocal Stiffness and Damping</i> .
CONFERENCE OR SYMPOSIUM PRESENTATIONS	[5] <i>Wave Propagation in an Elastic Lattice with Nonreciprocal Stiffness and Damping</i>, 19th International Congress on Artificial Materials for Novel Wave Phenomena (Metamaterials 2025), Amsterdam, The Netherlands, September 2025 (Scheduled). [4] <i>Characterizing a Parametric Oscillator using SINDy</i>, Mechanical Sciences 3rd Divisional Symposium - 2025 (theme: “Applications of AI/ML in Engineering”), Indian Institute of Science, Bengaluru, India, March 2025 (Poster presentation). [3] <i>Unraveling the behavior of waves in phononic time crystal</i>, Aerospace Research Students’ Symposium (AERES 2025), Department of Aerospace Engineering, Indian Institute of Science, Bengaluru, India, January 2025. [2] <i>Wave propagation in phononic time crystal</i>, 4th International Conference on Mechanics of Advanced Materials and Structures (ICMAMS 2024), Ramaiah Institute of Technology, Bengaluru, India, December 2024. [1] <i>Wave propagation in phononic time crystal</i>, Aerospace Research Students’ Symposium (AERES 2024), Department of Aerospace Engineering, Indian Institute of Science, Bengaluru, India, January 2024.
SEMINAR AND INVITED TALKS	Delivered a guest lecture in the course <i>Wave Propagation in Designed Materials</i>, offered by Prof. Rajesh Chaunsali at IISc Bangalore (Dec 2023).
MENTORING EXPERIENCE	- Mentor for Winter Intern at LEMS Lab, IISc Bangalore — delivered tutorials on nonlinear vibrations, with emphasis on parametric systems (Jan–June 2024). Teaching Assistant for the semester course AE 264 – Vibrations (Jan–May 2024).
LEADERSHIP AND SERVICE ACTIVITIES	Indian Institute of Science, Bengaluru, Karnataka, India - Served as Department Representative (DR) for the Department of Aerospace Engineering, appointed by the Student Council, IISc, during 2024–25. - Conducted a lab tour and experimental demonstration for Class 12 students from Rishi Valley School as part of the IISc–Agastya Science Enrichment Program 2024.
RELEVANT COURSES	Mathematics and Numerical Methods: Mathematical Methods for Aerospace Engineers (<i>AE 211</i>), Numerical Solution of Differential Equations (<i>DS 289</i>) Structural and Solid Mechanics: Vibrations (<i>AE 264</i>), Aeroelasticity (<i>AE 255</i>). Fluid Mechanics and Thermodynamics: Fluid Mechanics (<i>ME 201</i>), Fluid Turbulence (<i>ME 298</i>) Special Topics: Fluid–Structure Interaction (<i>AE 291</i>)
TECHNICAL SKILLS	Programming: C++, MATLAB, Python Documentation Tools: L^AT_EX, Microsoft Office Suite, ChatGPT Design and Visualization: Inkscape, CATIA V5 Simulation Software: COMSOL Multiphysics Experimental Tools: Laser Doppler Vibrometer (LDV), Data Acquisition Systems (DAQ), 3D Printing

PROFESSIONAL
MEMBERSHIPS

- Society for Industrial and Applied Mathematics (SIAM), 2021–Present
- American Physical Society (APS), 2023–2024

INTERESTS

Academic mentoring, music appreciation, keyboard playing, photography, spiritual reading, yoga, and trekking.

REFERENCES

Prof. Rajesh Chaunsali
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